

Week 11 Written Homework

● Graded

2 Days, 8 Hours Late

Student



Total Points

12 / 15 pts

Question 1

1

2 / 3 pts

+ 3 pts 3 pts: Meets Expectations -

The solutions provide a sufficient explanation of the methods used to solve the problem. (e.g. the solutions apply theorems appropriately, and justify any intermediary results).

There are no arithmetic errors nor minor gaps in logic.

✓ **+ 2 pts** 2pts: Approaches Expectations -

The solutions provide some explanations of the methods used to solve the problem. (e.g. the solutions generally apply theorems appropriately, but might not justify any intermediary results).

There may be minor errors or minor gaps in logic

+ 1 pt 1 pt: Revision Needed -

The solutions provide partial or incorrect explanations of the methods used to solve the problem. (e.g. the solutions may fail to verify key hypotheses; and/or the solutions may lack key details or intermediary results).
There are major errors or gaps in logic.

+ 0 pts Opts Not Assessable -

The solutions do not provide any logical explanations or justifications of the methods used to solve the problem.

There may be several major errors or gaps in logic.



incorrect part c

Question 2

2

2 / 3 pts

+ 3 pts 3 pts: Meets Expectations -

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+ 0 pts Opts Not Assessable -

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can still find convergence

Question 3

3

3 / 3 pts

✓ **+ 3 pts** 3 pts: Meets Expectations -

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+ 2 pts 2pts: Approaches Expectations -

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There may be minor errors or minor gaps in logic

+ 1 pt 1 pt: Revision Needed -

The solutions provide partial or incorrect explanations of the methods used to solve the problem. (e.g. the solutions may fail to verify key hypotheses; and/or the solutions may lack key details or intermediary results).
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+ 0 pts Opts Not Assessable -

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Question 4

4

2 / 3 pts

+ 3 pts 3 pts: Meets Expectations -

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✓ **+ 2 pts** 2pts: Approaches Expectations -

The solutions provide some explanations of the methods used to solve the problem. (e.g. the solutions generally apply theorems appropriately, but might not justify any intermediary results).
There may be minor errors or minor gaps in logic

+ 1 pt 1 pt: Revision Needed -

The solutions provide partial or incorrect explanations of the methods used to solve the problem. (e.g. the solutions may fail to verify key hypotheses; and/or the solutions may lack key details or intermediary results).
There are major errors or gaps in logic.

+ 0 pts Opts Not Assessable -

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3

once it simplifies to harmonic series it is no longer alternating so it diverges bc $p=1$

Question 5

5

3 / 3 pts

✓ **+ 3 pts** 3 pts: Meets Expectations -

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There are no arithmetic errors nor minor gaps in logic.

+ 2 pts 2pts: Approaches Expectations -

The solutions provide some explanations of the methods used to solve the problem. (e.g. the solutions generally apply theorems appropriately, but might not justify any intermediary results).
There may be minor errors or minor gaps in logic

+ 1 pt 1 pt: Revision Needed -

The solutions provide partial or incorrect explanations of the methods used to solve the problem. (e.g. the solutions may fail to verify key hypotheses; and/or the solutions may lack key details or intermediary results).
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Questions assigned to the following page: [2](#) and [1](#)

Math 102 Written HW 03/30

1) $\sum_{n=1}^{\infty} \frac{1}{7^n}$

a) $\frac{2801}{16807} = \sum_{n=1}^5 \frac{1}{7^n}$

b) ① positive ✓

② bounded ✓

③ always decreasing $\rightarrow 7^{-n} \rightarrow -\frac{\ln(7)}{7^x} \rightarrow$ decreasing!

~~$\int_6^{\infty} \frac{1}{7^n} \leq R_n \leq \int_5^{\infty} \frac{1}{7^n}$~~

~~$\left[\frac{-7^{-x}}{\ln(7)} \right]_6^{\infty} \leq R_n \leq \left[\frac{-7^{-x}}{\ln(7)} \right]_5^{\infty}$~~

~~$0 + \frac{7^{-6}}{\ln(7)} \leq R_n \leq 0 + \frac{7^{-5}}{\ln(7)} \approx 0.00003$~~

c) $R_n = S - S_n$ $\sum_{n=1}^{\infty} \left(\frac{1}{7}\right)^n \rightarrow \frac{a}{1-r} \rightarrow \frac{1}{1-\frac{1}{7}} \rightarrow \frac{1}{\frac{6}{7}} \rightarrow \frac{7}{6}$
 $R_5 = S - S_5$

$R_5 = \frac{7}{6} - \frac{2801}{16807}$

$S = \frac{7}{6}$

$R_5 = \frac{100843}{100842} \approx 1.00001$

2) $\sum_{n=1}^{\infty} \frac{\sin(n(\frac{\pi}{6}))}{1+n\sqrt{n}} \rightarrow N/A$ b/c ① $\sin(n(\frac{\pi}{6}))$ isn't alternating!

3) ~~$\sum_{n=1}^{\infty} -1 \cdot \frac{n^2 + e^{-n}}{2n^2 - n + 1} \rightarrow \lim_{n \rightarrow \infty} \frac{n^2}{e^n} \rightarrow 0$~~ but $|a_n|$ wouldn't work b/c -1^n

SC

Conditionally
Convergent

Questions assigned to the following page: [3](#), [5](#), and [4](#)

$$3) \frac{(-1)^n n^2 + e^{-n}}{2n^2 - n + 1} \rightarrow \lim_{n \rightarrow \infty} \frac{(-1)^n n^2}{2n^2} \rightarrow \lim_{n \rightarrow \infty} \frac{(-1)^n}{2} \rightarrow \text{diverges}$$

$$4) \underbrace{2-1}_{1} + \underbrace{1-\frac{1}{2}}_{\frac{1}{2}} + \underbrace{\frac{2}{3}-\frac{1}{3}}_{\frac{1}{3}} + \underbrace{\frac{2}{4}-\frac{1}{4}}_{\frac{1}{4}} \dots \quad (-1)^n \left(\frac{2}{n} - \frac{1}{n} \right)$$

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}, \text{ but, } \sum_{n=1}^{\infty} (-1)^n \left(\frac{1}{n} \right)$$

1) monotonically decreasing
 2) $\lim_{n \rightarrow \infty} \frac{1}{n} \rightarrow 0$
 by Alternating Series test $\rightarrow$ Converges

$$5) \int_n^{\infty} \frac{e^{-x}}{x} \leq .05$$

$$a) \frac{-1}{n} \rightarrow [-\ln(\ln 1)]_n^{\infty} \rightarrow \infty + \ln(n) \leq .05$$

$$e^{\ln(n)} = e^{-.05}$$

b) Alternating series only applies when $\frac{n \leq e^{-.05}}{n+1 \leq n} \rightarrow$ no long monotonically decreasing

$$\leq \sum_{n=1}^{\infty} (-1)^n \left(\frac{2}{n} - \frac{1}{n} \right) : -0.745$$

$$12: -0.658$$

$$15: ~~-0.705~~ -0.693$$

$$18: -0.662$$